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Adversarial Robustness Evaluation with Separation Index

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***Abstract*—The paper introduces a method to assess the robustness of deep neural networks against adversarial attacks. It employs a geometric-based separation metric called the Separation Index, which measures the distance between data points with distinct labels within the latent space of variational autoencoders utilized for classification tasks. The Separation Index quantifies the degree of data separation by comparing each data point with its neighboring data points. A higher value signifies greater separation between different classes, thus ensuring enhanced robustness. This approach yields dependable results when confronted with gradient-based adversarial attacks, including FGSM, R-FGSM, MI-FGSM, and PGD, under both white-box and black-box conditions.**

***Index terms*—** Separation Index, Robustness Evaluation, Variational Autoencoder

I. Introduction

Deep neural networks (DNNs) have made significant advancements in various domains, enhancing performance in tasks like computer vision, speech recognition, medical imaging, autonomous driving, and gaming. However, DNNs are vulnerable to adversarial attacks, where modified input samples cause misclassification with high confidence. Even subtle perturbations can greatly impact classifier output. Ensuring resilience to such attacks is crucial, particularly in safety-critical applications. Current research focuses on evaluating DNN robustness [1], developing attack algorithms [2], and defensive methods [3]. However, the development of robustness metrics for assessing DNNs against adversarial samples has received comparatively less attention, despite progress in attack and defense techniques.

The natural model accurately groups classes and exhibits clear decision boundaries with normal samples. However, its robustness diminishes when faced with adversarial samples, leading to blurred class groupings and decision boundaries. On the other hand, a robust adversarial-trained model maintains distinct decision boundaries and class groupings even in the presence of adversarial samples. To achieve robustness, patterns of the same labels should be closer together while patterns of different labels should be farther apart. This enhances the separation rate between feature points of different labels during feature extraction, minimizing the probability of misclassifications caused by adversarial examples crossing decision boundaries. The separation rate serves as a numerical index to evaluate the robustness of the deep neural network based on this principle.

We introduce the SI, a distance-based geometric factor that evaluates the robustness of deep artificial neural network (ANN) classifiers within the latent spaces of generative models. This paper focuses on shifting conventional performance indicators to the latent space, offering additional benefits for the probabilistic interpretation of this space. Our approach ensures that the data remains reliable even in the presence of adversarial perturbations. The main objective of this study is to establish and evaluate a performance metric for deep neural networks (DNNs) using the latent space of Variational Autoencoders (VAEs).

The remaining sections of the paper are organized as follows: In Section 2, we introduce the related works, while Section 3 provides the background. In Section 4, we formally introduce our evaluation metric, and in Section 5, we present the experimental results. Finally, Section 6 concludes the paper.

II. Related Works

The emergence of various adversarial attacks and defenses necessitates the need for a robustness metric to evaluate the performance of deep neural networks (DNNs) against adversarial samples. Unfortunately, there has been limited effort in developing such metrics. Currently, adversarial accuracy is commonly used as a metric to assess a model's robustness, which measures the model's ability to classify generated adversarial samples. However, adversarial accuracy lacks informative value and does not provide significant insights or knowledge about the model's behavior.

The radius of distortion, a crucial parameter in evaluating model robustness, represents the range of perturbation needed for effective adversarial examples. A resilient model with a larger perturbation radius indicates higher adversarial robustness, even with constant factors. Various distance metrics have been proposed to quantify perturbation, such as L_0, L_2, and L_∞. However, L_0 only counts affected pixels without considering disturbance magnitude, while L_2 can still be small despite minor changes. The L_∞ distance has been proposed as the optimal metric for robustness [4]. However, these metrics fail to account for various attack types and

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architectural variations. Adversarial frequency and severity statistics, suggested by [5], measure resilience through point-wise robustness. Adversarial frequency quantifies how frequently adversarial examples occur, dependent on ϵ and defined for the L norm. Adversarial severity assesses robustness independently of this hyperparameter. Point-wise robustness, measuring the proximity of adversarial examples to a given input x, serves as a local metric in this context. Certified robustness assessment approaches employ metrics or certifiers to establish verifiable lower bounds on the minimum adversarial distance for misclassification. Pioneering work [6], [7] has provided upper bounds and evaluation metrics considering task complexity. The Accuracy-Robustness Pareto Frontier (ARPF) [8] evaluates robustness in deep vision models against various attacks. Generative models in the latent space are proposed [9] to assess robustness. Optimum transport [10] investigates the discrepancy between state-of-the-art (SOTA) accuracy and optimal accuracy. Cross-Lipschitz Extreme Value for Network Robustness (CLEVER) [11] measures attack-agnostic robustness using Extreme Value Theory (EVT). However, CLEVER may overestimate perturbation size due to gradient masking, making it unreliable [12]. Enhancements propose the second-order CLEVER score with EVT and a twice-differentiable robustness guarantee, along with the use of Backward Pass Differentiable Approximation (BPDA) for non-differentiable input transformations [13].

Deep neural models can be evaluated for resilience against adversarial attacks based on their structure and parameters. Previous work [14] explored the resilience of deep models using topological concepts but did not provide specific robustness boundaries or estimations for neural networks. Another approach, DeepSafe [15], employed an automated data-driven method to identify the network's robust region through clustering labeled data and constraint solving. In a different approach, ROBY [16] introduced an attack-independent robustness metric that analyzes decision boundaries and feature space of models. By examining inter-class and intra-class statistic features, ROBY assesses the robustness of target neural network classifiers without relying on adversarial examples. A larger decision boundary distance, resulting from less overlap between distinct feature subspaces, indicates a more resilient deep model.

As many performance metrics, including accuracy, adversarial frequency, and severity, remain relevant when the original space is replaced with the latent one, the main approach used in this work to implement latent performance metrics is to shift existing performance metrics into the latent space.

III. BACKGROUND

A. Adversarial Attacks

To generate adversarial samples, numerous techniques have been developed. In this study, we focus on gradient-based methods, where adversarial examples can be created by taking perturbation steps in the opposite direction of the adversarial gradients.

The **Fast Gradient Sign Method (FGSM)** [17] is one of the earliest and most straightforward but highly powerful attacks. In equation 1.

$$x^{adv} = x + \epsilon_p.sign(\nabla_x J(x, y)) \tag{1}$$

Y represents the ground truth label of x, and the gradient of the loss function J (e.g., cross-entropy loss) with respect to the input image x is computed as the sign of the gradient, which is then multiplied by the magnitude ϵ_p to create perturbations. The parameter ϵ_p controls the similarity between adversarial instances and the original image by determining the magnitude of the perturbation.

A novel attack technique, known as the **Randomized Fast Gradient Sign Method (R-FGSM)** [18], introduces random Gaussian noise to an image before applying the FGSM algorithm to create perturbed images. The authors of this technique believed that defending against targeted attacks is more challenging than defending against untargeted attacks. Therefore, they conducted targeted adversarial attacks against FGSM-based approaches. For the experiments, the authors randomly selected target labels from the classes, excluding the true class.

The **Momentum Iterative Fast Gradient Sign Method (MI-FGSM)** [19], is an optimization technique that incorporates momentum. The authors argue that the use of momentum helps stabilize update directions for perturbations and allows for the evasion of weak local maxima.

$$g_{n+1} = \mu.g_n + \frac{\nabla_x J(x_n^{adv}, y)}{\|\nabla_x J(x_n^{adv}, y)\|_2} \tag{2}$$

$$x_{n+1}^{adv} = x_n^{adv} + \alpha.sign(g_{n+1}) \tag{3}$$

The **Projected Gradient Descent Method (PGD)** [20] attack is proposed for highly constrained optimization. The PGD attack performs multiple iterations to find an adversarial sample during the generation process. Initially, it starts searching for an adversarial instance at a pre-defined location within an l_p norm ball randomly. By utilizing the noisy starting point in adversarial training, a more powerful attack is achieved compared to previous iterative methods, thereby enhancing the effectiveness of their defense.

B. Variational Autoencoder

An observation in the latent space can be probabilistically described using the variational autoencoder (VAE) model [21]. Consequently, the VAE formulates the encoder to describe a probability distribution for each latent attribute instead of constructing an encoder that provides a single deterministic value for each latent state feature. This approach allows for a probability distribution to represent each latent property for a specific input. The encoder is not involved in the generation of output, and the decoder generates an input variable by randomly selecting values from a distribution of latent states. To handle the non-differentiability of such a random sampling procedure, a reparameterization approach is employed during the training process. The VAE minimizes the lower bound of the $\log p(x)$ for negative evidence as follows:

$$
\begin{aligned}
&L_{VAE} \\
&= E_{q(z|x)}[\log \frac{p(x|z)p(z)}{q(z|\hat{x})}] \\
&= E_{q(z|x)}[\log p(x|z) + D_{KL}(q(z|\hat{x})p(z))]
\end{aligned} \tag{4}
$$

Where the first term represents the Reconstruction Loss and the second term describes the KL-Regularizer.

IV. A Metric for Robustness Evaluation

A. Separation Index

The First Order SI, as described in [22], quantifies the degree of separation between data points with different labels. In this method, the separation of data is determined by comparing each data point with its neighboring data points and assessing whether they belong to the same class. The index ranges from "zero" to "one", where a value closer to "one" indicates a higher level of separation between classes. Conversely, a value closer to "zero" suggests that data from different classes are mixed.

In this algorithm, the first step is to calculate the Euclidean distance between all pairs of data points, regardless of their labels. Based on these distances, the closest data point to each data point is identified, and a comparison is made between the original data point and its closest neighbor, taking into account their labels. To compute the separation index(SI), the SI variable is initially set to zero. Then, it is checked whether the label of the desired data point and its closest neighbor is the same. If they are, one unit is added to the SI variable; otherwise, it remains unchanged. This process is repeated for each data point, and the total value is calculated. Finally, this total value is divided by the total number of data points to normalize the result between zero and one.

$$SI(\{z^q\}_{q=1}^{Q}, \{l^q\}_{q=1}^{Q}) = \frac{1}{Q}\sum_{k=1}^{Q}\varphi(l^q - l^{near}) \tag{5}$$

$$\varphi(v) = \begin{cases} 1 & v = 0 \\ 0 & v \neq 0 \end{cases}, q_{near} = \arg\min_{h} \|z^q - z^h\|^2 \tag{6}$$

Where $\{z^q\}_{q=1}^{Q}$ denotes the dataset with its labels and $\varphi(.)$ denotes the Kronecker delta function.

In fact, the number of data points with labels that are identical to those of their closest neighbors (neighbors with minimum distances) is counted by the first-order SI. The rationale behind this metric is that a robust algorithm ensures that its classification result remains consistent within a closed neighborhood for any sample taken from its input distribution. A higher SI between different classes across all input samples indicates a greater level of robustness in the DNN.

B. Distance Matrix

In this method, Euclidean distances are computed between all data points, regardless of their labels. To avoid computational inefficiency, an alternate approach is employed. This

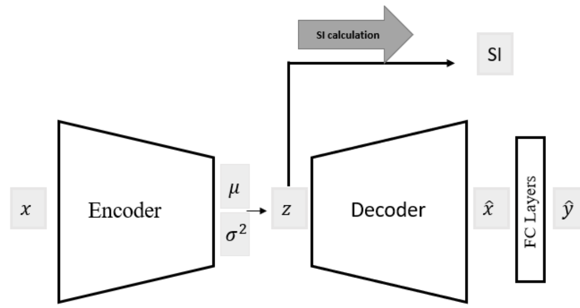

Fig. 1: Scheme of Using SI for VAE-Based Classifier Method [23]

method utilizes a distance matrix (D) that encompasses pairwise distances between the data points. Equation 7 calculates the distance matrix D. Matrices P and G are determined using equation 8, while matrices p and X are obtained through equation 9.

$$D_{n\times n} = P^t + P - 2G \tag{7}$$

$$P_{n\times n} = \begin{bmatrix} p & p & \dots & p \end{bmatrix}_{n\times n}, G_{n\times n} = XX^T \tag{8}$$

$$p_{n\times 1} = \begin{bmatrix} \|x_1\|_2^2 \\ \|x_2\|_2^2 \\ \vdots \\ \|x_n\|_2^2 \end{bmatrix}_{n\times n}, \quad X_{n\times m} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{9}$$

In the equations, x_i represents a feature vector of data i with dimensions $1 \times m$. The distance matrix D stores the distances between data points, with each entry representing the distance between the i-th and j-th data points.

V. Model

To determine the SI of the input's latent distribution, which contains valuable information, we utilize the Variational Autoencoder (VAE) model. In our work, contrary to the typical testing procedure of VAE, we retain the encoder to extract robust features. Through the VAE, we project the original data onto a lower-dimensional latent space to estimate the probability density and capture the original manifold.

To enhance the robustness of the feature extractor and achieve robust features in the latent space, we employ the method proposed in [23]. This method combines the classifier and the VAE into a single network. By incorporating a classifier during the model training, we obtain a resilient data manifold after the encoding process. By incorporating classifier optimization in the VAE loss function, we introduce a constraint that encourages not only a continuous latent space but also sufficient separation between different classes for easy discrimination. The loss function for this method is shown in Equation 10.

$$L_{total}(\theta, \phi, \beta) = L_{\beta VAE} + \alpha.L_{classification} \tag{10}$$

Where $L_{classification}$ is Cross-Entropy loss and $L_{\beta VAE}$ is shown in 11:

$$
\begin{aligned}
&L_{\beta VAE} \\
&\quad = E_{q(z|x)}[\log \frac{p(x|z)p(z)}{q(z|\hat{x})}] \\
&\quad = E_{q(z|x)}[\log p(x|z) + \beta D_{KL}(q(z|\hat{x})p(z))]
\end{aligned} \tag{11}
$$

As we aim is to assess the SI in the latent space as a metric for robustness, it is crucial to achieve the most resilient features in that space. By doing so, we can ensure that the SI criterion, which quantifies the distance between different classes, is a suitable measure for evaluating the network's resilience level. Therefore, we employ the method proposed in [23] and evaluate the SI criterion in the latent space, as illustrated in Figure 1. Evaluating the SI index in the latent space can provide valuable insights into various aspects, such as the network's robustness, attack potency, the richness of latent space features, and more.

VI. Experiments

To assess the efficacy of our method in evaluating defense against adversarial attacks, we conduct tests involving both white-box and black-box attack scenarios. We generate adversarial examples using gradient-based attack methods, including FGSM, Rand-FGSM, MI-FGSM, and PGD. Our experiments are conducted on the MNIST [24] and CIFAR-10 [25] datasets, which are widely recognized as standard benchmark datasets in the field of machine learning. For single-channel data, such as MNIST, we utilize a VAE structure comprising multiple convolutional layers, building upon our previous work [23], and The ResNet18 architecture was employed for three-channel data.

As described in [23], the maximum magnitude of the adversarial perturbation is limited to 0.3. To optimize the model, we use the Adam optimizer with an initial learning rate of 0.001, which exponentially decreases to around 0.00001. The hyperparameters for MNIST are set as follows: hidden space size = 128, α = 0.85, and β = 1. The maximum magnitude of the perturbation for the CIFAR-10 data, for single-step attacks, is limited to 0.3, and for other attacks, it is limited to 8/255. The corresponding hyperparameters for CIFAR-10 are as follows: hidden space size = 256, α = 0.5, and β = 1.

A. Separation Index Evaluation

As this index examines the separability value of the data from each other, it is expected that the SI value grows dramatically compared to the raw adversarial data after applying the defense method. Table I compares the SI metric of adversarial samples before and after feeding them into the network in the latent space. The features in the latent space have a higher SI value than the original adversarial samples. This outcome confirms the effectiveness of SI metrics in evaluating robustness.

When examining the results presented in Table I and comparing them with previous research [23] that primarily employs accuracy as a metric for evaluating robustness, a significant

TABLE I: Comparison of SI for latent space of our model and natural space of images for different attacks for MNIST dataset

	Mnist		Cifar-10	
Space / Attack	Natural	Latent	Natural	Latent
FGSM	0.81	0.98	0.24	0.63
R-FGSM	0.8	0.97	0.22	0.54
MI-FGSM	0.78	0.92	0.27	0.43
PGD	0.71	0.91	0.24	0.41

and noteworthy correlation between Accuracy and the SI emerges. This correlation suggests that the SI metric holds promise as a valuable alternative for assessing robustness.

One of the key advantages of utilizing the SI metric over accuracy lies in its ability to eliminate the requirement for intricate knowledge regarding the internal workings and architectural characteristics of the models being evaluated. Traditional accuracy-based evaluations often necessitate a deep understanding of the model's architecture and parameters, which can be resource-intensive and time-consuming. In contrast, the SI metric simplifies the evaluation process by relying solely on input and output data, thereby bypassing the need for detailed model information.

Moreover, the SI metric offers an added benefit in terms of providing an intuitive understanding of the data. By quantifying the degree of separation between data points, the SI metric provides valuable insights into the distribution and characteristics of the data under consideration. This intuitive understanding can be instrumental in making informed decisions when selecting the most suitable model from a range of alternatives.

Furthermore, the SI metric streamlines the search process for identifying the most robust model among several options. By leveraging the SI metric, researchers and practitioners can effectively narrow down the pool of potential models and focus their efforts on those that exhibit higher separation indices. This targeted approach reduces the time and computational resources required for evaluating and comparing different models, ultimately expediting the search for the most robust model for a given task.

B. Visualization

To analyze the structure of the data distribution and assess the generalizability of the model, we visualize the latent space. To achieve this, we employ the Uniform Manifold Approximation and Projection (UMAP) [26] technique for dimensional reduction. This technique helps visualize the high-dimensional feature vectors extracted by the trained model and plot the decision boundary in a two-dimensional space. Figure 2 presents various visualization results for comparison. The left figures display the clustering of adversarial examples under different attacks, while the right figures depict the clustering of the latent space using the method introduced for adversarial data. The decision boundaries in the latent space exhibit greater clarity compared to those of the natural adversarial

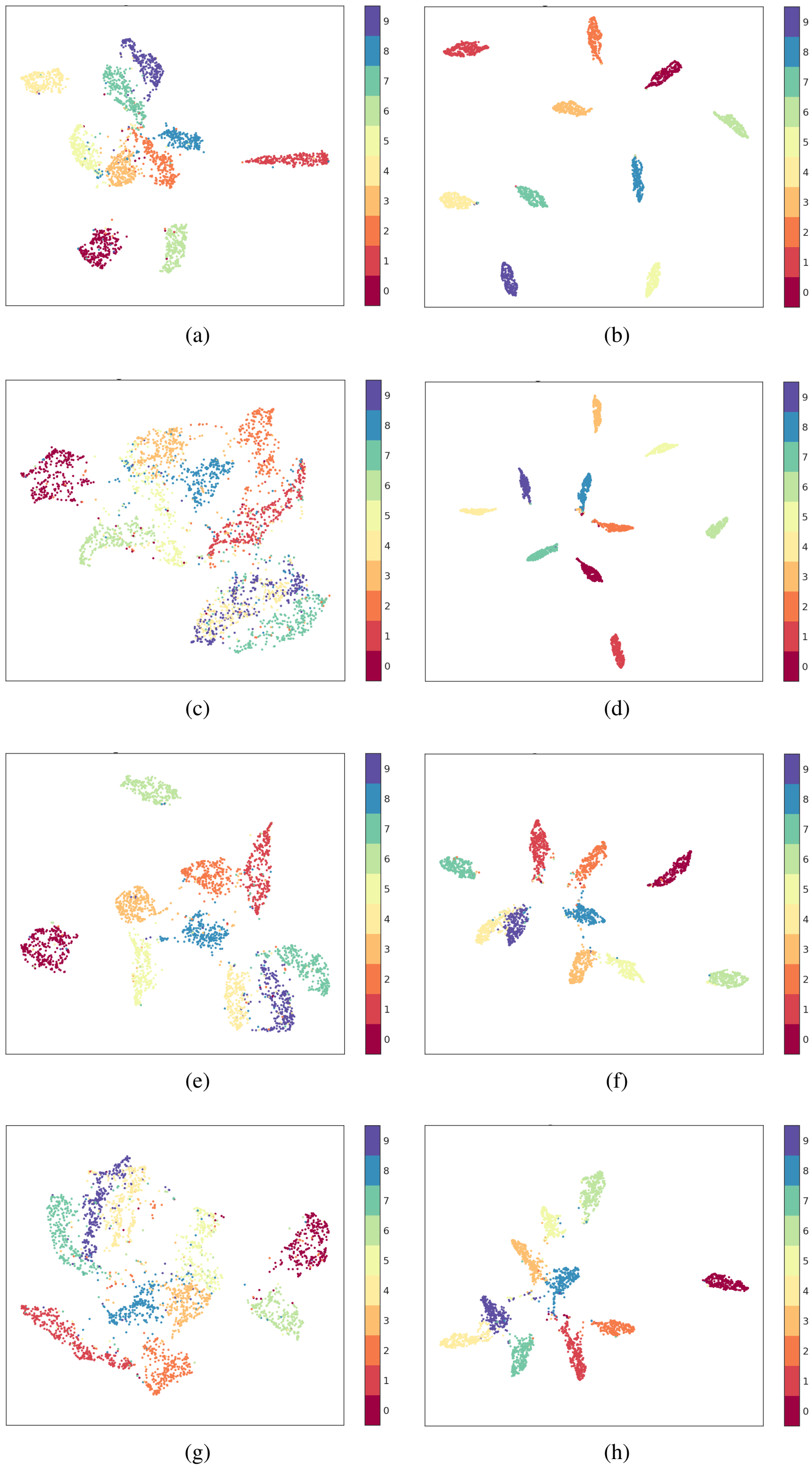

Fig. 2: Clustering plots of MNIST dataset in natural space and latent space under differnt adversarial attacks. (a), (c), (e), and (g) are respectively related to the natural space of adversarial examples under FGSM, R-FGSM, MI-FGSM, and PGD attacks. (b), (d), (f), and (h) are respectively related to latent variables of the proposed method in [23] FGSM, R-FGSM, MI-FGSM, and PGD attacks.

samples. Additionally, the robust features in the latent space demonstrate higher data compactness within the same class and more distinct separation across classes, leading to reduced overlap among all classes.

VII. Conclusion

The paper presents a novel approach to assess the robustness of networks against adversarial attacks using the geometric SI. In this study, we propose the utilization of this index in the latent space, which represents essential features. The results demonstrate that the index exhibits lower values for adversarial raw data but performs better in the VAE latent space. This index can serve as an alternative to accuracy for evaluating network robustness since accuracy calculations require more time. Moreover, we aim to incorporate this index into the loss function to enhance the network's robustness during adversarial training.

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